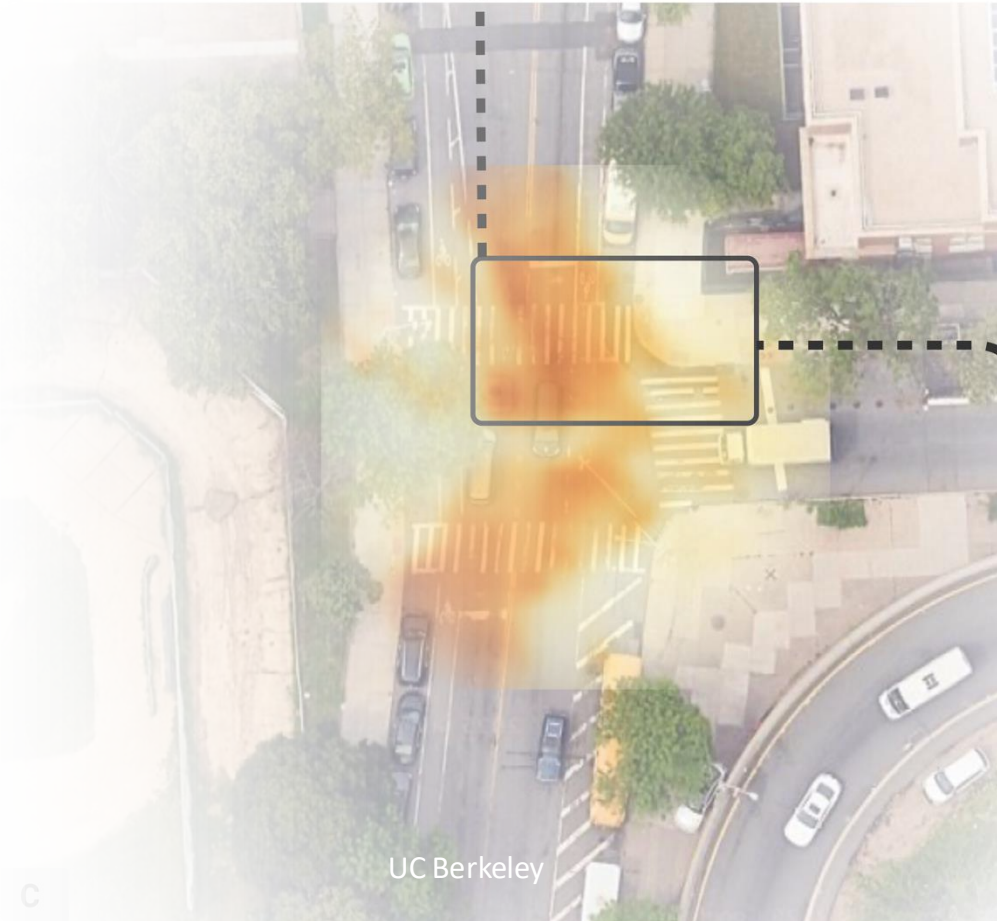
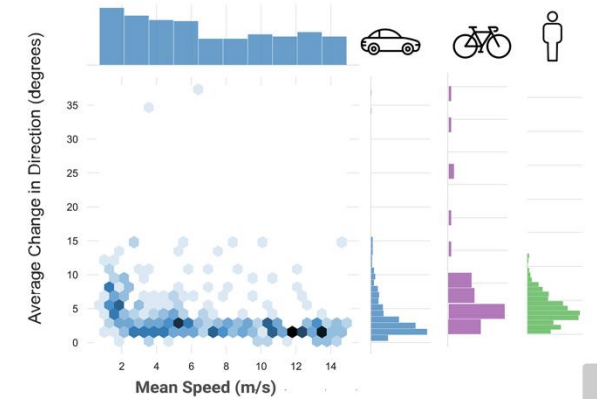
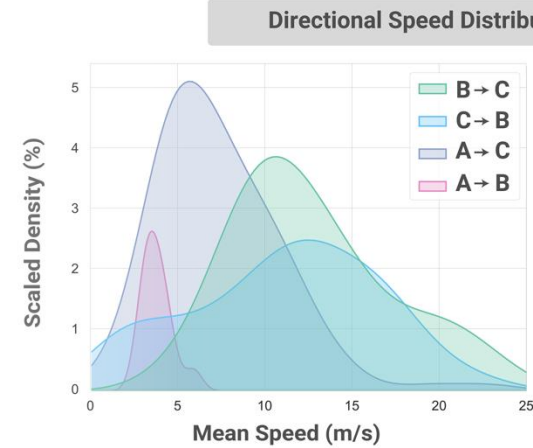
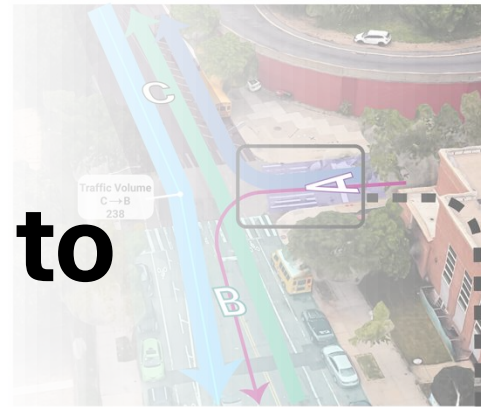


Introduction to Urban Data Analytics

Fall 2026

Maryam Hosseini

City & Regional Planning



B

D

Overview



Your Questions

Q - How to Find Teammates?

A- Checkout the Ed Discussion.

If you could not find it till Sep 7, we can we will help and will find you a team!

Q- How much we'll be using python. Each homework assignment? Labs?

A- A lot! All assignments, most of the lab and classroom notebooks, and your final project will be in Python.

Q- How technical can we get with our projects?

A- As technical as you want! At minimum, you should use some of the methods we cover in class, but you can always go beyond that. Bring in your own domain knowledge and impress us. You will also get feedback on your project along the way, so we will help you maintain the right balance.

Your Questions

Q – I have prior coding experience, which final project should I choose?

A- Track B.

Q- how theoretical is this class versus examining real world examples & applications?

A- We will use theory to frame the questions, but starting around Weeks 4–5, we will work with real world urban data through the rest of the semester.

Q- Should readings be completed before lecture or after lecture?

A- Before lectures.

Q- What are the lab topics?

A- Checkout the course schedule!

Your Questions

Q - Can we use other datasets we find that are not listed on course website?

A - Yes!

Introduction to Urban Data Analytics



[Final Project](#) / Datasets

Example Datasets for the Final Project

You can use the curated sources below to identify data for your final project. The lists emphasize open licensing, clear documentation, and spatial usability. All items are intended to be directly usable in Python/GIS workflows. You can use any other publicly available datasets.

Final Project

Where can I read about Final Project?

On the website, click **Final Project** in the **top navigation bar**.

Syllabus

Due Dates

Final Project

Office Hours

Final Project Due Dates

1. Group Formation

Sep 9 – 12:00 pm

2. Proposal and Lightning Talk Video

Sep 21 – 12:00 pm

3. Midterm Submission

Nov 2 – 11:59 pm

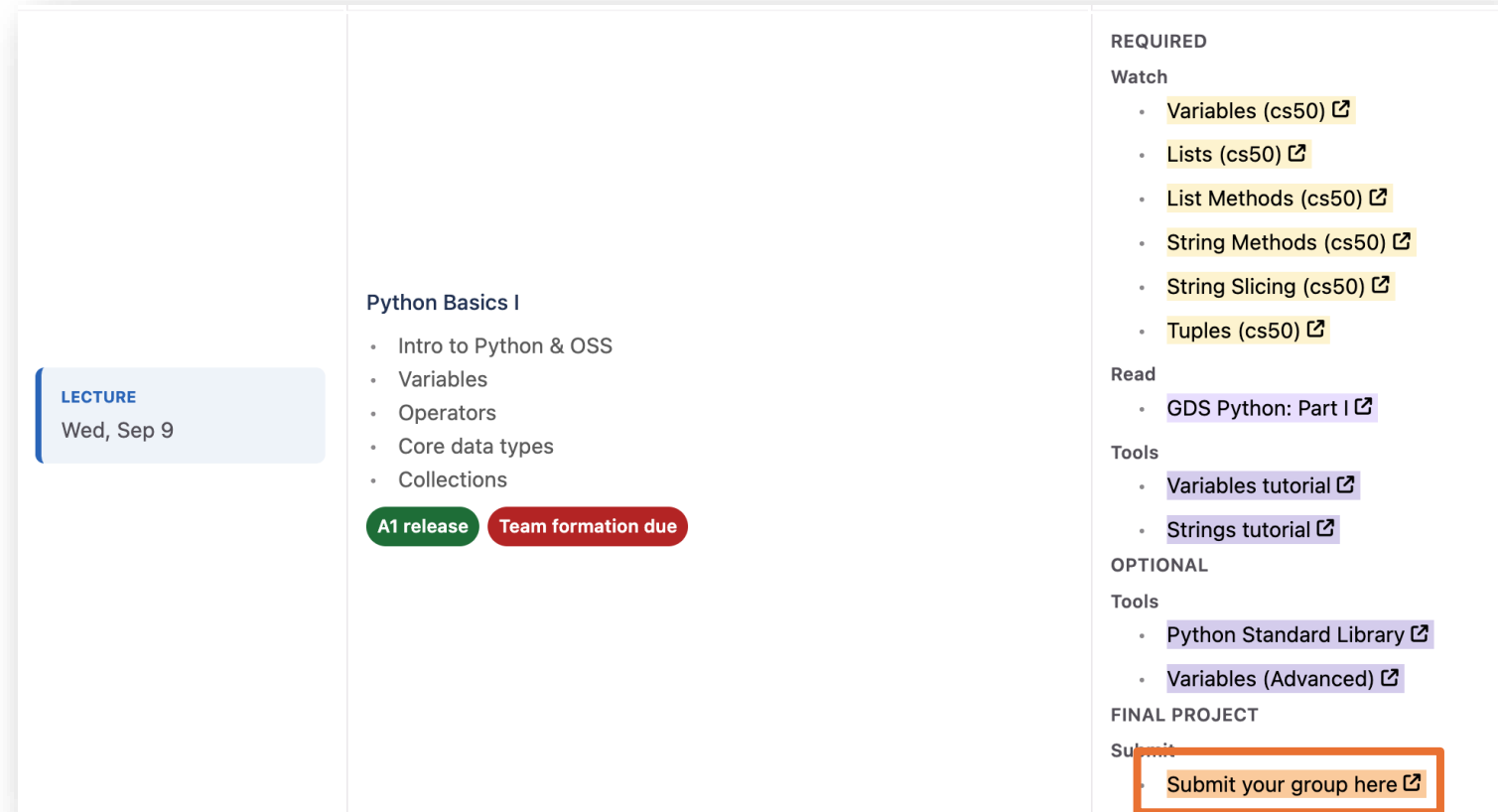
4. Final Project Submission

Dec 15 – 11:59 pm

Final Project

Group Formation

Due Sep 9 – 12:00 pm



The screenshot shows a course page layout. On the left, a blue box labeled 'LECTURE' indicates the date 'Wed, Sep 9'. The main content area is titled 'Python Basics I' and lists topics: 'Intro to Python & OSS', 'Variables', 'Operators', 'Core data types', and 'Collections'. Below this list are two buttons: 'A1 release' (green) and 'Team formation due' (red). On the right side, under the 'REQUIRED' section, there is a 'Watch' list with links to 'Variables (cs50)', 'Lists (cs50)', 'List Methods (cs50)', 'String Methods (cs50)', 'String Slicing (cs50)', and 'Tuples (cs50)'. Below that is a 'Read' section with 'GDS Python: Part I'. Under 'Tools', there are links for 'Variables tutorial' and 'Strings tutorial'. The 'OPTIONAL' section includes 'Python Standard Library' and 'Variables (Advanced)'. At the bottom, under 'FINAL PROJECT', there is a 'Submit' section with a link 'Submit your group here' which is highlighted by an orange box and an orange arrow from the text 'Where to submit?' below the screenshot.

LECTURE
Wed, Sep 9

Python Basics I

- Intro to Python & OSS
- Variables
- Operators
- Core data types
- Collections

A1 release Team formation due

REQUIRED

Watch

- Variables (cs50)
- Lists (cs50)
- List Methods (cs50)
- String Methods (cs50)
- String Slicing (cs50)
- Tuples (cs50)

Read

- GDS Python: Part I

Tools

- Variables tutorial
- Strings tutorial

OPTIONAL

Tools

- Python Standard Library
- Variables (Advanced)

FINAL PROJECT

Submit

- Submit your group here

Where to submit?

Infrastructure

- **Resources**

Course website

- **Coding**

JupyterHub on DataHub

- Jupyter Notebook

- **Submission**

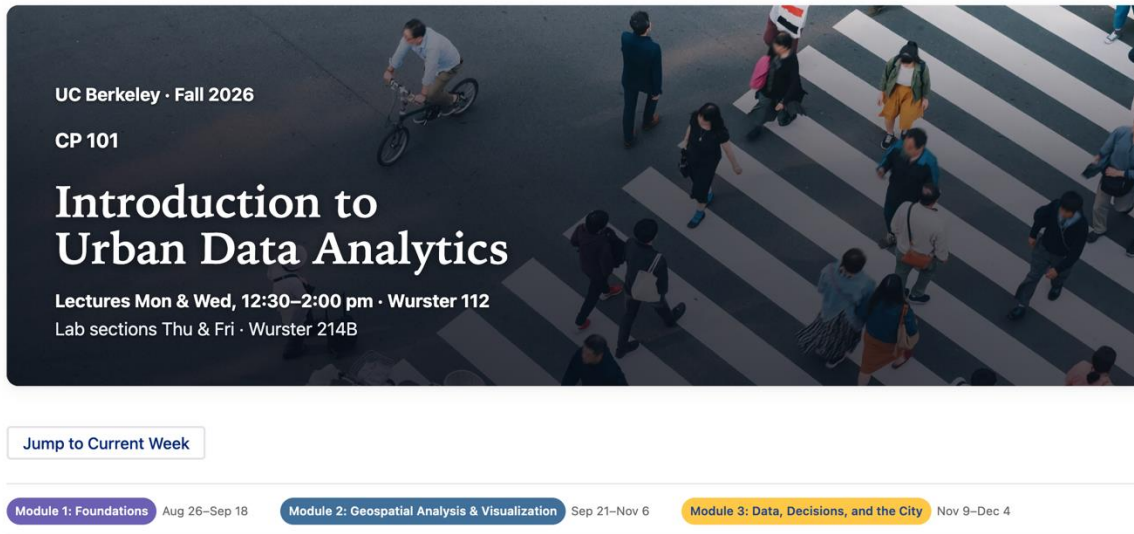
Gradescope

- **Questions, Discussion & GSI Support**

Ed Discussion

Course Website

<https://ucb-cp101.github.io/fa26/>



Main source of information about this course.
The site will be updated frequently.

- **Bookmark It:** Add the website to your bookmarks and check it regularly for announcements and materials.
- **Readings and Resources:** Lecture slides, reading and videos and extra materials
- **Assignments, Labs and Class Activities**
- **Syllabus** and Course Policies and

Geospatial Data



Projection, Vector & Raster Data

Where are you?

A location is meaningful with a reference frame

- 106
- Wurster 106
- Cal, Wurster Hall
- Cal
- At Berkeley
- In the Bay Area
- In California
- In the U.S.

**(37.8703°, -122.2595°) in WGS84 /
EPSG:4326**



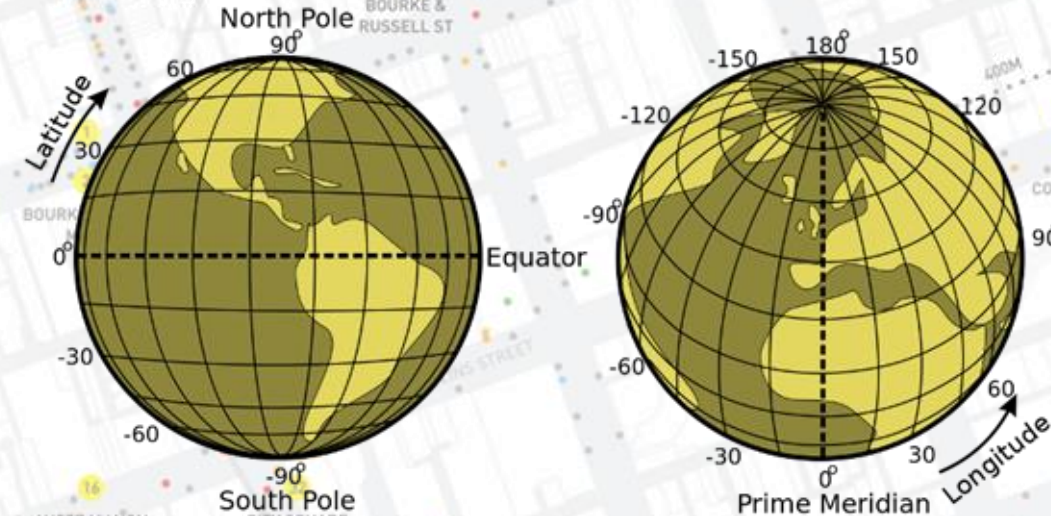
Spatial Data

- **Spatial** means “about space/geometry.”
- It can describe location, shape, distance, topology, or 3D structure in **any coordinate system**; Earth-based or not.

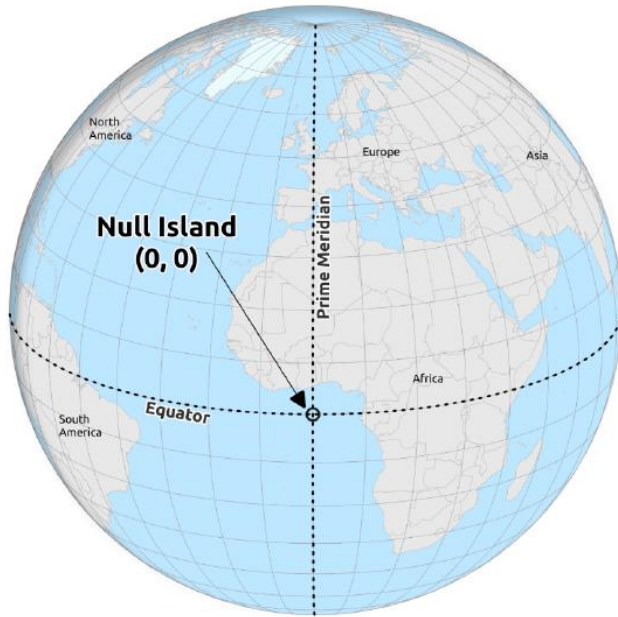


Geospatial Data

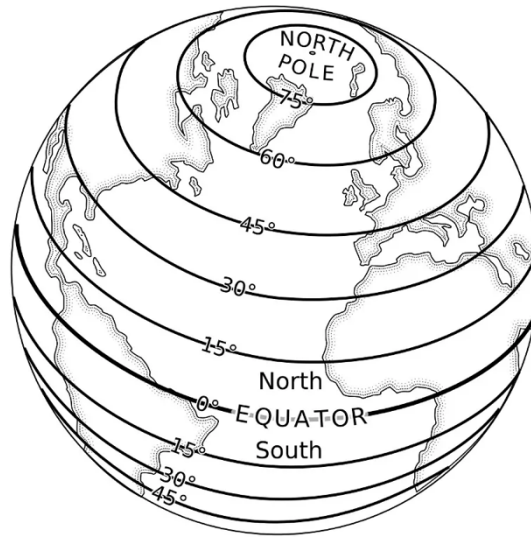
- Geospatial data is data with Earth's surface as the reference frame: **data tied to locations on Earth's surface**
- How can we uniquely identify any location on Earth?
- By using a geographic coordinate grid
- Any point on Earth can be specified with a pair of coordinates:
 - latitude (north–south)
 - longitude (east–west)



Latitude and Longitude

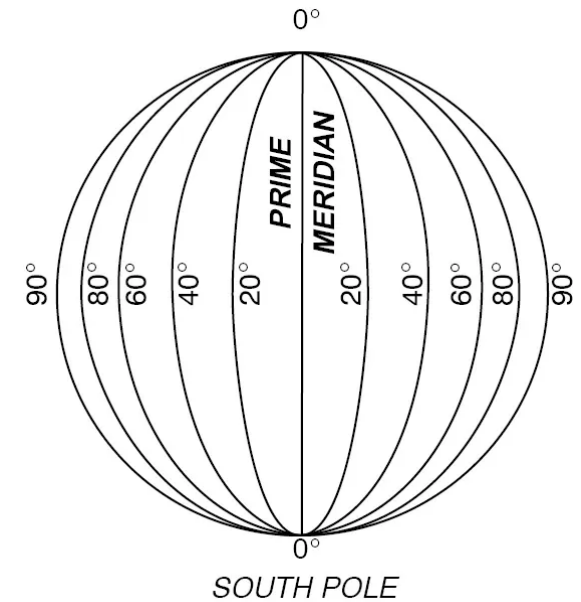


The origin (0,0), is where **Prime Meridian** and **Equator** intersect



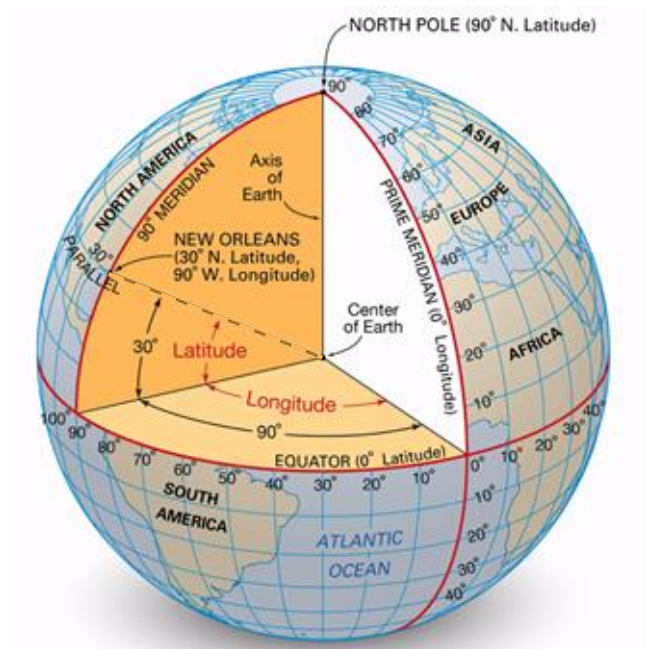
Lines of Latitude

The lines that run east and west each have a constant latitude value and are called **parallels**.



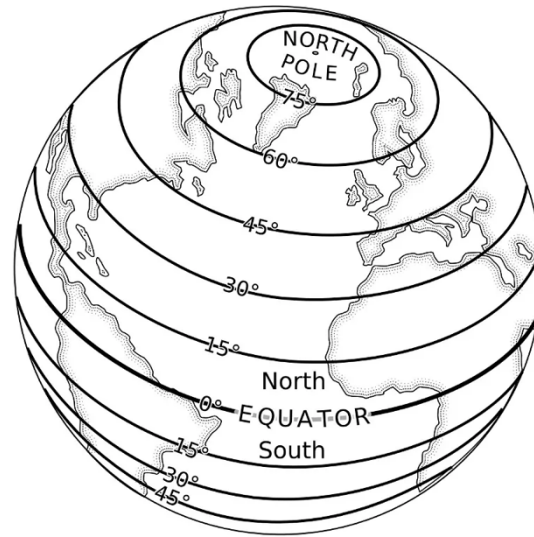
Lines of Longitude

The lines that run north and south each have a constant longitude value and are called **meridians**.



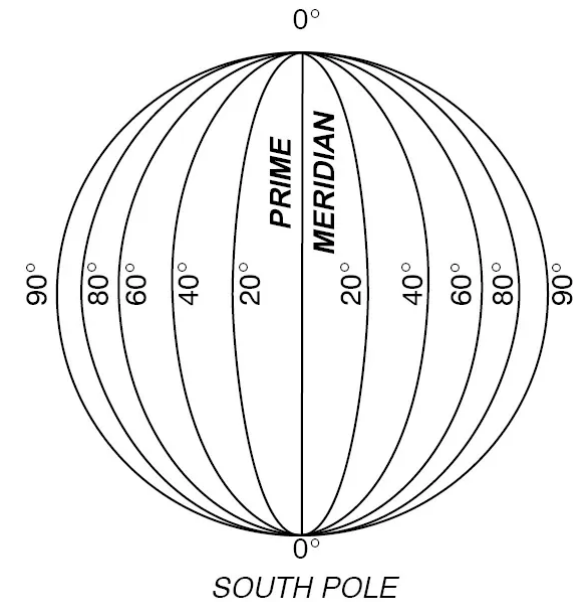
© Encyclopædia Britannica, Inc.

The origin (0,0), is where **Prime Meridian** and **Equator** intersect



Lines of Latitude

The lines that run east and west each have a constant latitude value and are called **parallels**.



Lines of Longitude

The lines that run north and south each have a constant longitude value and are called **meridians**.

From 3D to 2D

All maps lie flat; therefore, all maps lie.

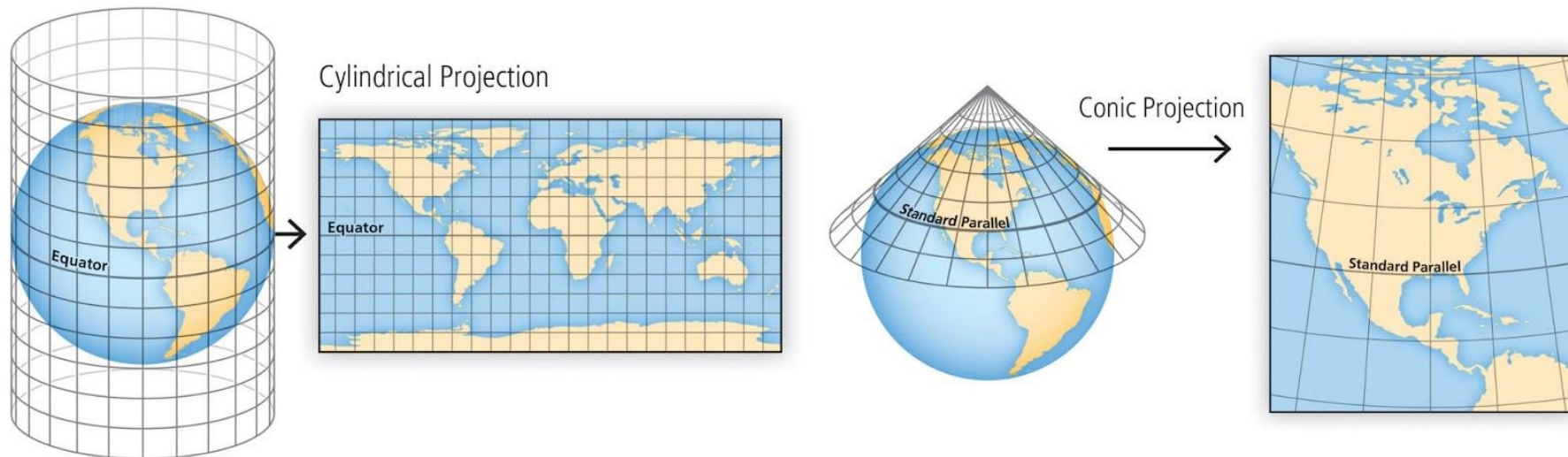
An old cartographic adage
Cited in (Hendrickson, 1994)



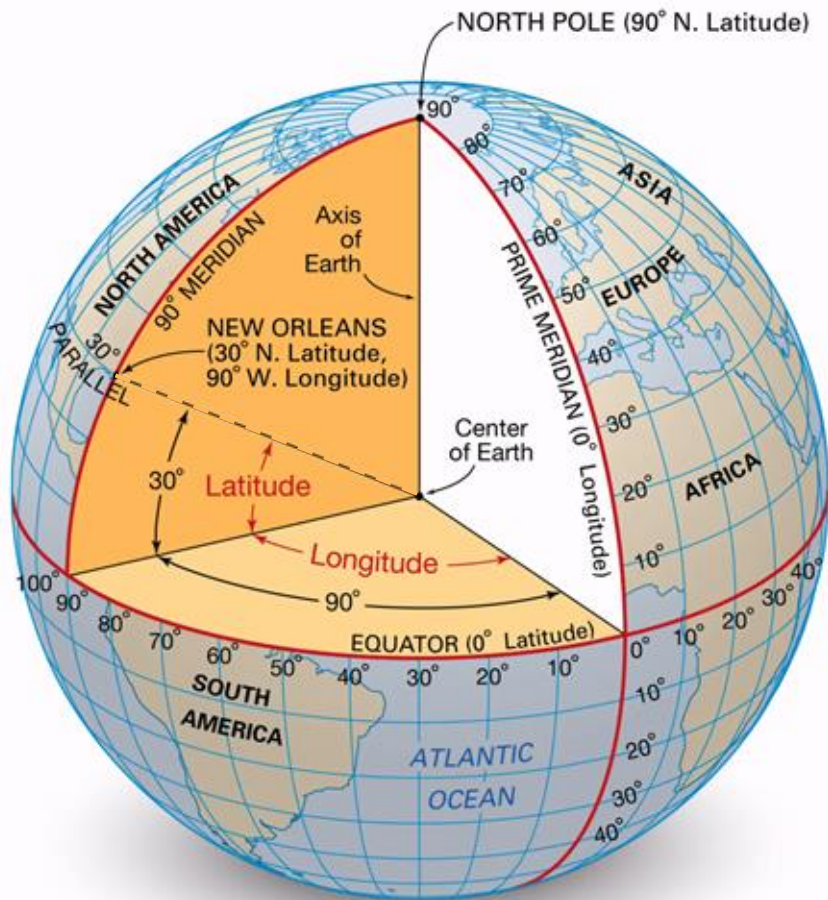
Vox

Projection

- **Projection:** mathematical transformations that convert the 3D, curved surface of the Earth into a 2D, flat map
- Projection is a key concept of cartography, the art and science of making maps



Projection



© Encyclopædia Britannica, Inc.



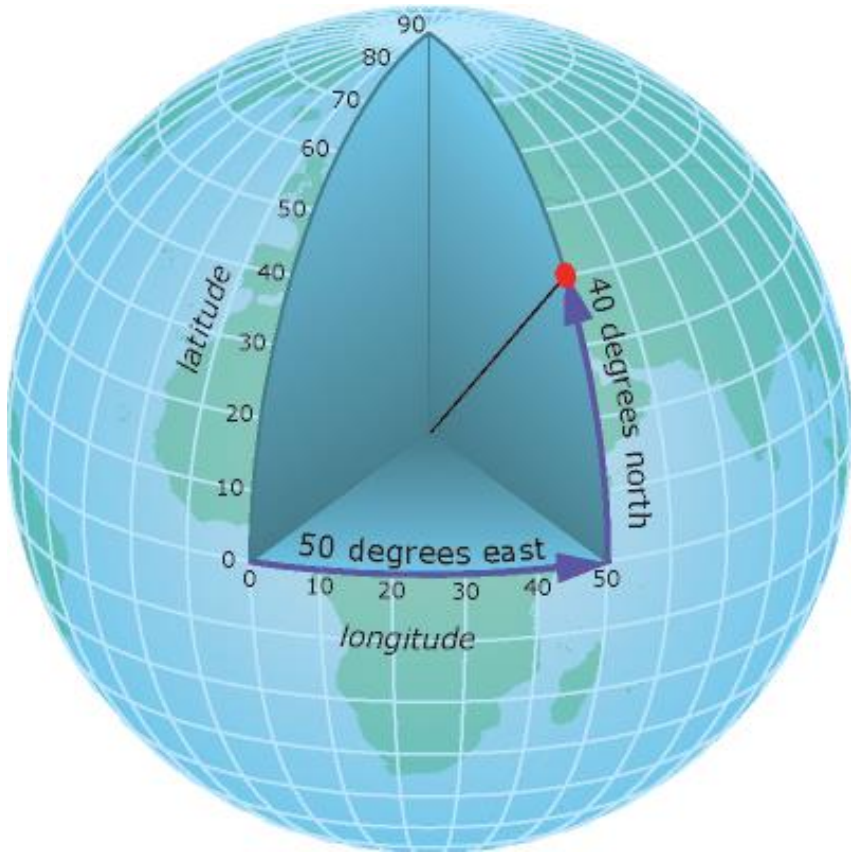
(lon, lat)



(x, y)

An angle as the unit of measurement does not tell you about distance or area. It is computationally difficult to compute distances and areas in that system.

Projection



(lon, lat)



(x, y)

An angle as the unit of measurement does not tell you about distance or area. It is computationally difficult to compute distances and areas in that system.

Coordinate Reference System (CRS)

A **coordinate reference system** (CRS) defines how the two-dimensional, projected map relates to real places on the earth.



Spherical Model
**Geographic
Coordinate System**
(GCS)



Flat Model
**Projected
Coordinate System**
(PCS)

Why bother with different CRS?

All projections distort some combination of:

- Shape
- Size
- Distance
- Direction
- Combination of these

The best projection depends on the use of the map.

World Mercator projection with country going to true size



@neilrkaye

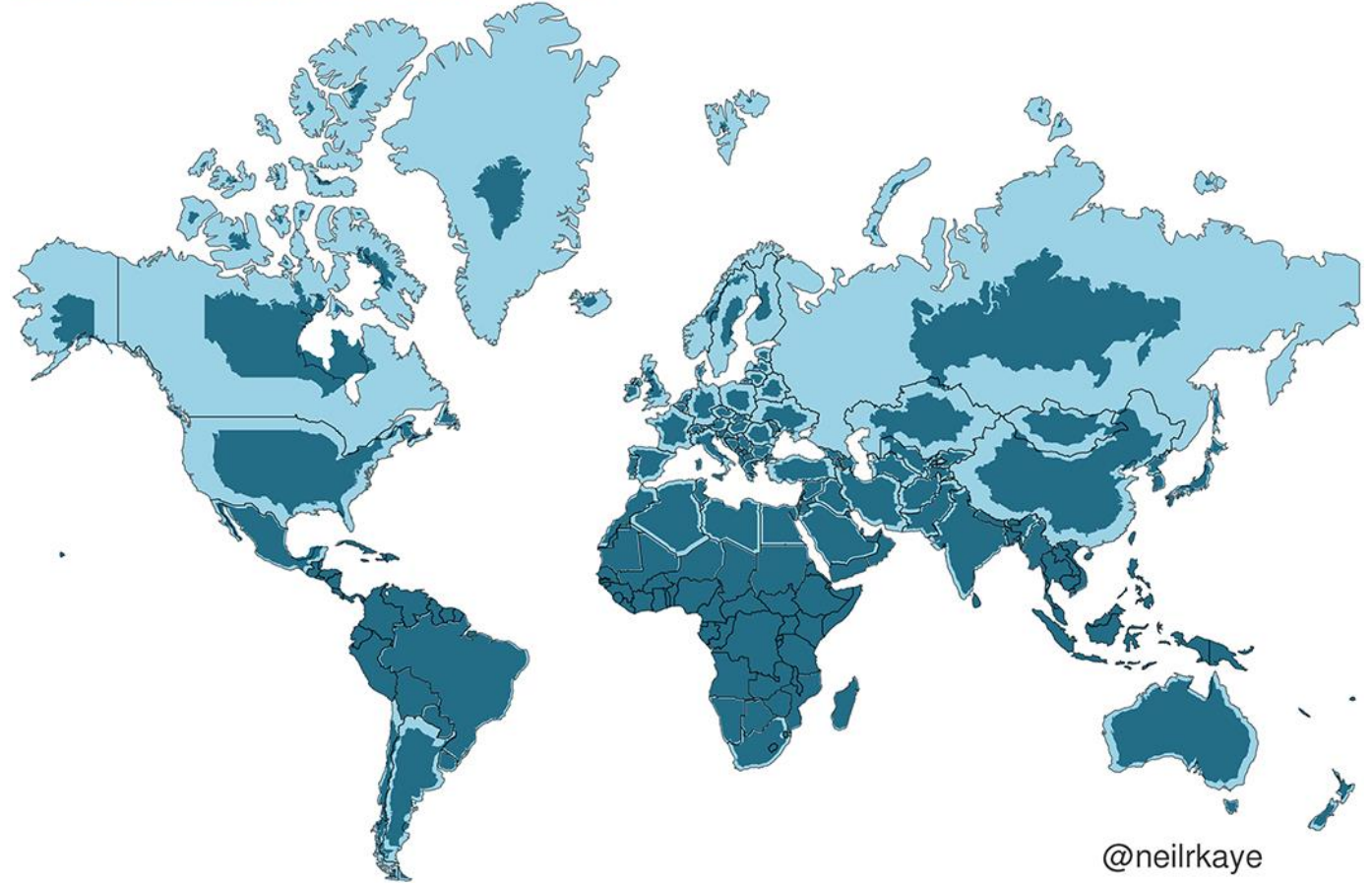
Why bother with different CRS?

All projections distort some combination of:

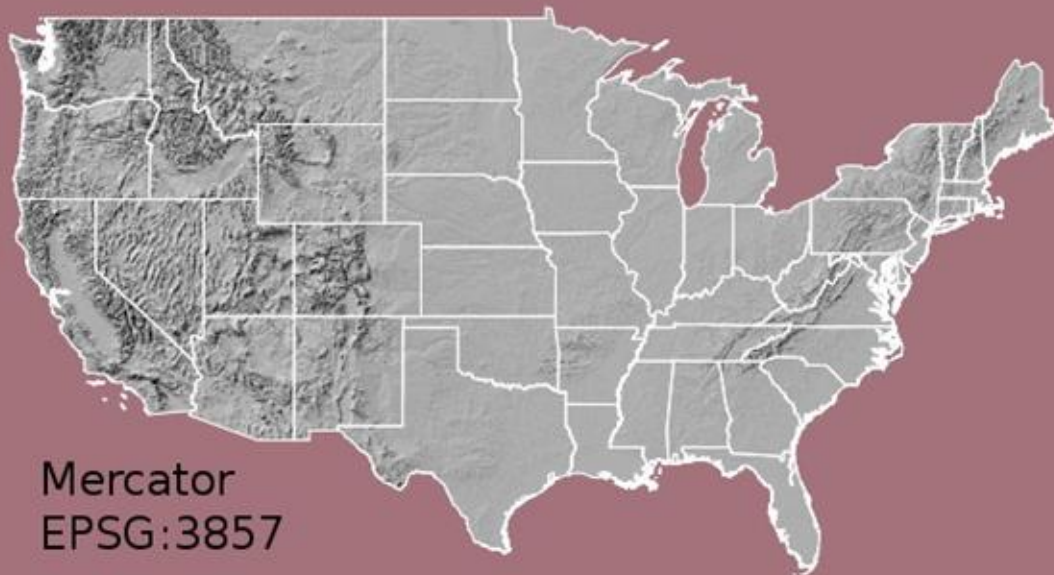
- Shape
- Size
- Distance
- Direction
- Combination of these

The best projection depends on the use of the map.

MERCATOR PROJECTION VS THE TRUE SIZE OF COUNTRIES



@neilrkaye



Mercator
EPSG:3857



UTM Zone 11N
EPSG:2955



U.S. National Atlas
Equal Area
EPSG:2163



WGS 84
EPSG:4326

Coordinate Reference System (CRS)



CRS connects data to the Earth's surface using a mathematical model

Geographic CRS:

- Uses latitude and longitude (angular degrees).
- Defines positions using latitude and longitude (angles) measured from the equator and prime meridian **on a specified datum/ellipsoid (e.g., WGS 84)**.
- Common example: **EPSG:4326 (WGS84)** (World Geodetic System (WGS) used in GPS).

Projected CRS

- Converts Earth's curved surface into a **2D map** with X, Y coordinates.
- Uses **meters or feet** instead of degrees
- Introduces **distortions**: different regions adopt their own CRS
- **Common example: EPSG:3857 (Web Mercator)** widely used for web mapping, **Significant area distortion near the poles.**

Class activity

Understanding Projections (25 minutes)

Go to **Course website** and under week 2, ACTIVITY, open [Map Projection](#)

- Follow the instructions
- Discuss in groups
- Share your answers (at least one volunteer team)

ACTIVITY

- [Map Projection](#) ↗

Geographic Information System (GIS)

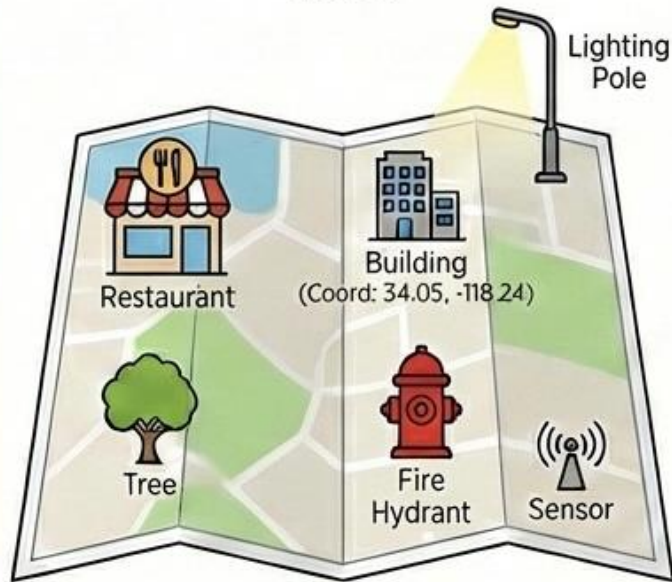
“A system for
capturing, storing, checking, integrating,
manipulating, analyzing & displaying
spatial data”

(British Department of the Environment, 1987)

GEOSPATIAL DATA CONCEPTS

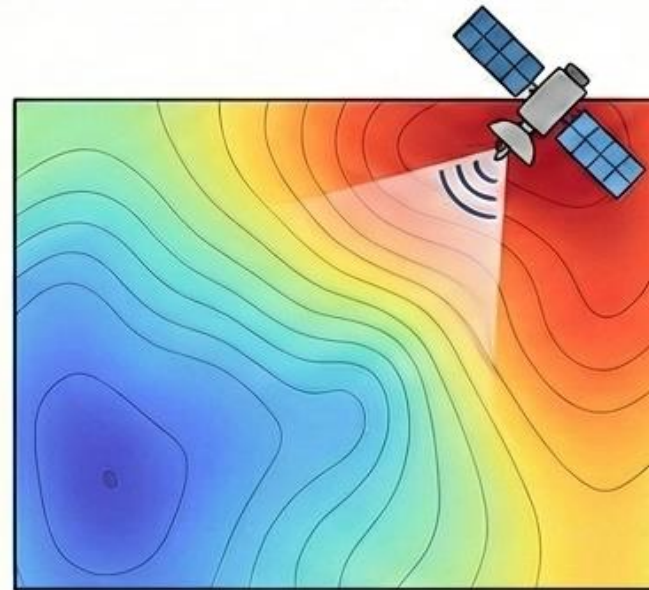
OBJECTS

Discrete entities that occupy a specific position in space and time.



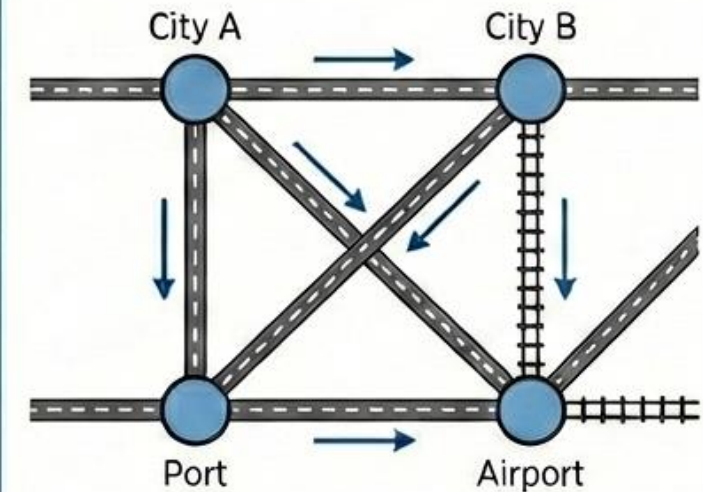
FIELDS

Continuous surfaces that could, in theory, be measured at any location in space and time.



NETWORKS

A set of connections between objects or between positions in a field.

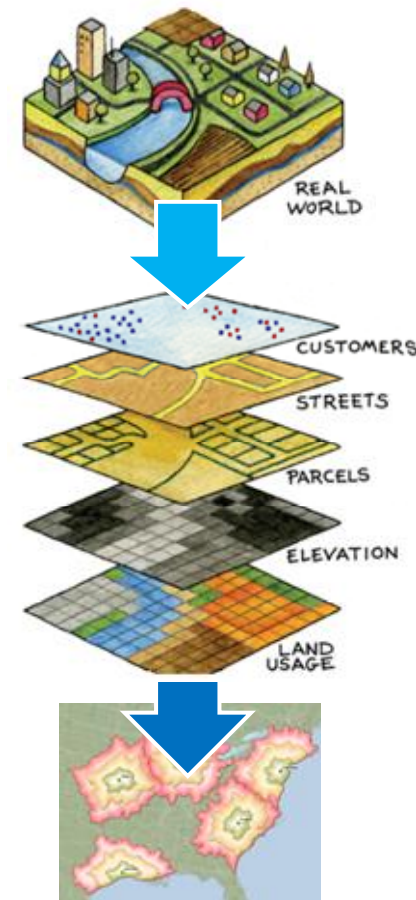


Theoretical Overview

GIS recreates real world spatial data

as digitized themed data “layers”
(e.g. locations, boundaries, infrastructure,
land use/cover)

assembled in any combination &
overlaid for analysis



Vector Data



- Vectors consist of vertices (x, y) that define spatial shapes.
- Their arrangement determines the type:
 - **Points:** Single x, y coordinate (e.g., POI sites)
 - **Lines:** Two or more connected vertices forming a continuous line (e.g., roads)
 - **Polygons:** Three or more vertices that are connected and closed (e.g., boundaries,)

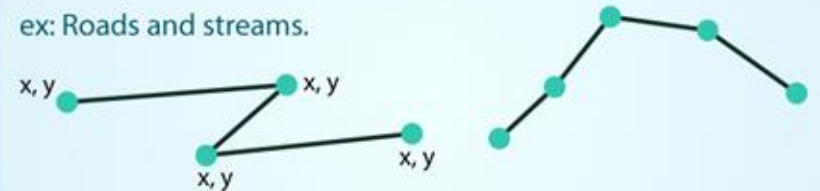
POINTS: Individual x, y locations.

ex: Center point of plot locations, tower locations, sampling locations.



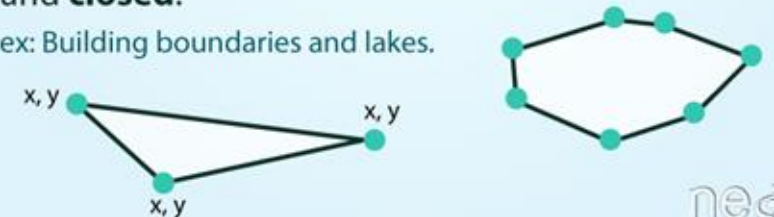
LINES: Composed of many (at least 2) vertices, or points, that are connected.

ex: Roads and streams.



POLYGONS: 3 or more vertices that are connected and **closed**.

ex: Building boundaries and lakes.

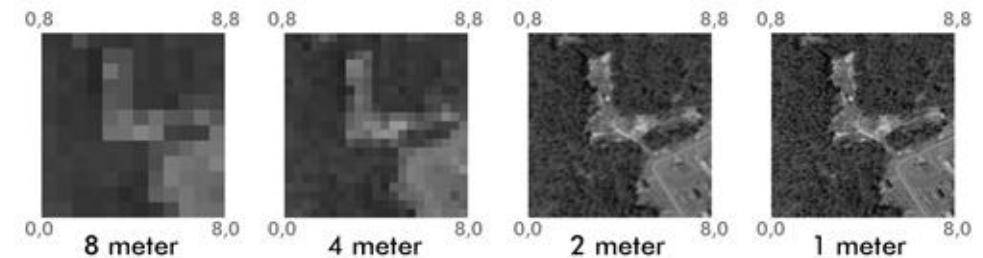
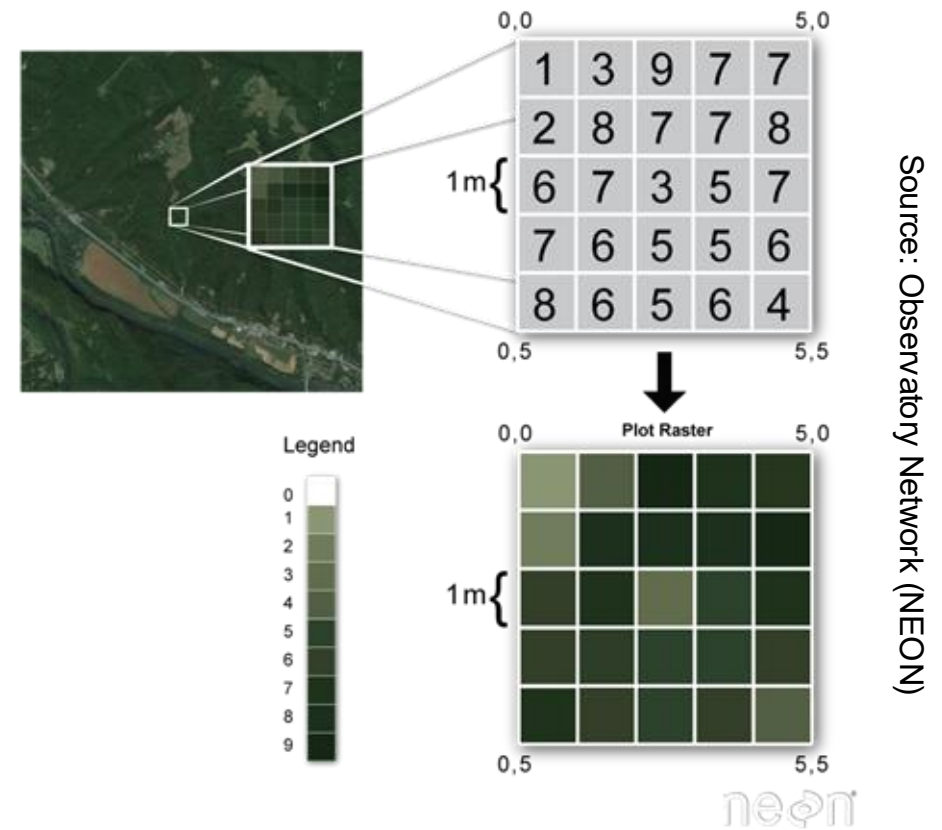


neon

Image Source: National Ecological Observatory Network (NEON)

Raster Data

- Raster data is made up of pixels (tiny squares), each representing a specific area on the Earth's surface.
- The size of each pixel, known as the Ground Sampling Distance (GSD), determines the resolution of the raster data.
- A smaller GSD means higher resolution and more detail, while a larger GSD means lower resolution and less detail.
- Each pixel has a value that represents information, such as color, elevation, temperature.
- Raster data includes aerial photographs, digital elevation models, and scanned maps.



Raster over the same extent,
4 different resolutions

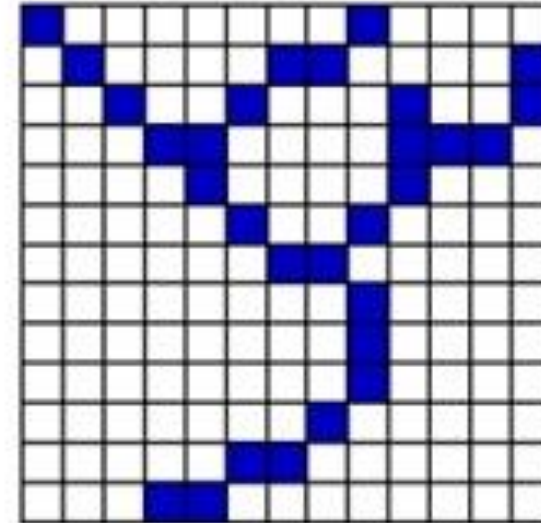
Data Types: Vector & Raster

Vectors are composed of coordinates



Vector

Rasters are composed of pixels



Raster

Data Types: Vector & Raster

Vectors are composed of coordinates and are best suited to manmade features with defined locations and boundaries.

Rasters are composed of pixels and are best suited to variables, usually environmental, that change over surfaces such as temperature, precipitation, or elevation.

